**Normal forms exercises**

1. Consider a relation R with the schema R(A, B, C, D, E, F) with a set of functional dependencies

F = {AB → C, BC → AD, D → E, CF → B}

Find all candidate keys of relation R

2. The relation R {A, B, C, E, F, G, H} contains only atomic values.

F = {CH → G, A → BC, B → CFH, E → A, F → EG} is a set of functional dependencies for R. Find all candidate keys of relation R.

3. Consider a relation R {A, B, C, D, E, F, G, H} and a set of functional dependencies:

S = {A → CD, ACF → G, AD → BEF, BCG → D, CF → AH, CH → G, D → B, H → DEG}.

Find all keys for R

4. Find all keys of R = (A,B,C,D,E) given a set of functional dependencies:

F = { A → BC, CD → E, B → D, E → A }

5. Given the table

| OID | O_Date | CID | C_Name | C_State | PID | P_Desc | P_Price | Qty |
|---|---|---|---|---|---|---|---|---|
| 1006 | 10/24/09 | 2 | Apex | NC | 7 | Table | 800 | 1 |
| 1006 | 10/24/09 | 2 | Apex | NC | 5 | Desk | 325 | 1 |
| 1006 | 10/24/09 | 2 | Apex | NC | 4 | Chair | 200 | 5 |
| 1007 | 10/25/09 | 6 | Acme | GA | 11 | Dresser | 500 | 4 |
| 1007 | 10/25/09 | 6 | Acme | GA | 4 | Chair | 200 | 6 |

Find all functional dependencies and Keys for the table above.

6. Find functional dependencies and keys on following table

| SID | CID | S_name | C_name | Grade | Faculty | F_phone |
|---|---|---|---|---|---|---|
| 1 | IS318 | Adams | Database | A | Howser | 60192 |
| 1 | IS301 | Adams | Program | B | Langley | 45869 |
| 2 | IS318 | Jones | Database | A | Howser | 60192 |
| 3 | IS318 | Smith | Database | B | Howser | 60192 |
| 4 | IS301 | Baker | Program | A | Langley | 45869 |
| 4 | IS318 | Baker | Database | B | Howser | 6019 |

7. Consider relation: R = (A, B, C, D, E) with the set of functional dependencies:

F = {CE → D, D → B, C → A}

a. Find all candidate keys.
b. Identify the best normal form that R satisfies (1NF, 2NF, 3NF).
c. If the relation is not in 3NF, decompose it until it becomes 3NF.
    - At each step, identify a new relation, decompose and re-compute the keys and the normal forms they satisfy.

8. Consider relation: R = (A, B, C, D, E, F, G, H, I, J) with the set of functional dependencies:

F = {AB → C, A → DE, B → F, F → GH, D → IJ}

a. Find all candidate keys.
b. Identify the best normal form that R satisfies (1NF, 2NF, 3NF).
c. If the relation is not in 3NF, decompose it until it becomes 3NF.
    - At each step, identify a new relation, decompose and re-compute the keys and the normal forms they satisfy.

9. Consider relation: R = {A, B, C, D, E, F, G, H, I, J} and the set of functional dependencies:

F = {AB → C, BD → EF, AD → GH, A → I, H → J}

a. Find all candidate keys.
b. Identify the best normal form that R satisfies (1NF, 2NF, 3NF).
c. If the relation is not in 3NF, decompose it until it becomes 3NF.
    - At each step, identify a new relation, decompose and re-compute the keys and the normal forms they satisfy.

10. Given following table

| staffNo | dentistName | patientNo | patientName | appointment | | surgeryNo |
|---|---|---|---|---|---|---|
| | | | | date | time | |
| S1011 | Tony Smith | P100 | Gillian White | 12-Aug-03 | 10.00 | S10 |
| S1011 | Tony Smith | P105 | Jill Bell | 13-Aug-03 | 12.00 | S15 |
| S1024 | Helen Pearson | P108 | Ian MacKay | 12-Sept-03 | 10.00 | S10 |
| S1024 | Helen Pearson | P108 | Ian MacKay | 14-Sept-03 | 10.00 | S10 |
| S1032 | Robin Plevin | P105 | Jill Bell | 14-Oct-03 | 16.30 | S15 |
| S1032 | Robin Plevin | P110 | John Walker | 15-Oct-03 | 18.00 | S13 |

Identify all functional dependencies, then define the highest Normal form for this table. If the table is not in 3NF, decompose the table into a set of 3NF tables.

11. Given following table

| NIN | contractNo | hoursPerWeek | eName | hotelNo | hotelLocation |
|---|---|---|---|---|---|
| 113567WD | C1024 | 16 | John Smith | H25 | Edinburgh |
| 234111XA | C1024 | 24 | Diane Hocine | H25 | Edinburgh |
| 712670YD | C1025 | 28 | Sarah White | H4 | Glasgow |
| 113567WD | C1025 | 16 | John Smith | H4 | Glasgow |

Identify all functional dependencies, then define the highest Normal form for this table. If the table is not in 3NF, decompose the table into a set of 3NF tables.